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Also corrected here: HT7833 absolute max input is 8V, not 28V — battery chemistry matters (spec §2.1). GPIO7 (J3 pin3) is NOT the stock "D7" castellation (spec §2.2).

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ESP32-C6-MINI-1-N8
 8MB flash - PCB antenna
 RISC-V @160MHz, Wi-Fi 6 / BLE5

do not touch

CAMERA / SPI INTER

J2 — to doorlock MCU
(Tuya keep-alive, 5-pin)

TXD_MCU	1 TXD_MCU
RXD_MCU	2 RXD_MCU
SRDY	3 SRDY
MRDY	4 MRDY
	5 GND

dedicated GND pin — keeps UART clean during motor current spikes

Design basis

Modified Seed XIAO ESP32-C6 R1.0
Antenna, crystal, SW1/SW2 unchanged

2.1 HT7833 input rating (corrected)

Datasheet max Vin is 8V, not 28V.
Alkaline/NiMH 4xAA (~4.8-6.6V) is fine;
primary lithium AA (~7.2V+ fresh) eats
nearly all remaining margin. Specify
chemistry on packaging.

2.2 GPIO7 is not "D7"

Stock D7 castellation = GPIO17
(U0RXD default). GPIO7 is a different
pin (MTDO/JTAG alias). Both work fine
via GPIO matrix, but D7 must be left
floating (not GPIO7) since GPIO17 is
now committed to TXD_MCU.

2.3 Connector count (see banner)

17 pins across J1-J4 will not fit this
outline. Full analysis in spec.

2.4 I2C traded away

D4/D5 (GPIO22/23) are the stock I2C
pins (SDA/SCL). Repurposing them for
UART1 removes I2C from this board.

Removed vs. stock XIAO

USB-C connector — omitted
SGM40567 charger IC — omitted
SGM6029C buck reg — omitted
(pads may remain, unpopulated)

Symbol key

 ground

+3.3V
 +3.3V tap

net label — same color+name=same net, no drawn wire

J3 — programming (UART0)

1 3.3V

2 TXD_DBG

3 RXD_DBG

4 GND

SW1

EN

Reset - pulls EN low

SW2

GPIO9 / BOOT

Boot - pulls GPIO9 low

Net-label tags connect to the matching color+name tags in the Core Module section (no drawn cross-wire lines)

Diagram illustrating the SPI interface connections for the J4 header:

- 1 SCK
- 2 MOSI
- 3 MISO
- 4 CS
- 5 3.3V
- 6 GND

The connections are shown as follows:

- SCK pin connected to 1 SCK
- MOSI pin connected to 2 MOSI
- MISO pin connected to 3 MISO
- CS pin connected to 4 CS
- +3.3V pin connected to 5 3.3V
- GND pin connected to 6 GND

ESP32-C6-MINI-1-N8 Doorlock + Camera Baseboard — Engineering Specification

Rev B — modified Seeed XIAO ESP32-C6 R1.0, 6V (4xAA) battery · companion sheet: schematic (page 1)

Before this goes to Seeed: three items below need a decision from you before this is buyable/manufacturable — the connector pin count doesn't fit the outline (§2.3), the LDO's real input ceiling is lower than assumed (§2.1), and one GPIO label in your spec pointed at the wrong physical pin (§2.2). Everything else in your brief checked out against Seeed's published schematic.

1. Overview

This is a from-scratch redesign of the previous baseboard, now explicitly built as a modified Seeed XIAO ESP32-C6 (R1.0) rather than a bespoke board using the bare module. The antenna, crystal, and both strap buttons stay untouched. USB-C, the charger IC, and the buck converter are removed in favor of a single linear regulator run from 4xAA cells, and a fourth connector (J4) adds a dedicated SPI bus for a Tuya camera MCU alongside the existing doorlock MCU link.

- **Core:** ESP32-C6-MINI-1-N8, unchanged antenna/crystal/strap circuitry from stock XIAO ESP32-C6
- **Power:** 6V nominal (4xAA, ~4.8-6.6V alkaline/NiMH range), HT7833-3.3 LDO to 3.3V
- **Doorlock MCU link (J2):** dedicated UART1 (GPIO17/GPIO22) + SRDY/MRDY handshake + dedicated GND
- **Camera MCU link (J4):** dedicated SPI bus (GPIO18/19/20/21)
- **Programming (J3):** UART0 (GPIO16/GPIO7) via external USB-UART adapter

2. Design review notes — please read before layout

2.1 HT7833 input ceiling is 8V, not 28V

Holtek's HT7833 datasheet lists an 8V input rating, not 28V — confirmed against three independent distributor listings (JLCPCB, LCSC, Holtek's own datasheet). Typical quiescent current is ~4-7 μ A (datasheet says 4 μ A typ, some distributor listings show 7 μ A), and dropout is roughly 500mV at 500mA for the 3.3V part. This changes the safety-margin story:

- 4xAA **alkaline or NiMH** tops out around 6.4-6.6V fresh — comfortably under 8V, this is fine.
- 4xAA **primary lithium** cells (e.g. Energizer Ultimate Lithium) run ~1.8V/cell fresh — about 7.2V pack voltage, leaving under 1V of margin to the LDO's absolute maximum. Not recommended without re-verifying against the exact cells you'll ship, and I would not rely on this margin in a security product.
- The original D2 TVS (SMBJ5.0A, 5V standoff) would have falsely clamped during *normal* 6V operation — it's resized to SMBJ8.5A in this revision. A TVS in this voltage class protects against ESD/surge transients; it does not substitute for choosing the right battery chemistry, since a device that clamps below ~8V would also clamp during normal operation.
- Recommendation: specify alkaline or NiMH only on the packaging/manual, and add an explicit warning against primary lithium AA cells.

2.2 GPIO7 is not the stock “D7” pad

Checked against Seeed's published XIAO ESP32-C6 schematic: the D7 castellation is wired to **GPIO17** (the default U0RXD IO_MUX pin), not GPIO7. GPIO7 is a different chip pin (the MTDO/JTAG alias) that isn't broken out to any D-pad on the stock board at all.

Your J3 table assigns GPIO7 to UART0 RX and labels it “D7” — GPIO7 will work fine electrically (ESP32-C6's UART is fully matrix-routable), but it is **not** the same net as the D7 castellation, so the “D7” label is misleading for anyone doing bring-up with a probe. Separately, your J2 table already uses GPIO17 for TXD_MCU, which is the *real* D7 net — so D7 needs to be added to the non-functional/floating castellation list, or the stock trace from GPIO17 to the D7 pad needs to be deliberately cut, otherwise TXD_MCU is quietly also exposed on an unlabeled edge pad.

Both fixes are reflected on the schematic: GPIO7's tag is annotated “(not the stock D7 pad)”, and D7 is added to the floating-pad list in §5 below.

2.3 Connector pin count does not fit 21.0 x 17.5mm

J1 (2) + J2 (5) + J3 (4) + J4 (6) = 17 discrete pin positions across four separate connector bodies, all specified for the same bottom edge of a 21mm-wide board.

Scenario	Math	Result
At 2.54mm pitch (J1 as specified)	17 pins x 2.54mm	= 43.2mm of pin pitch alone — double the board width
At 1.27mm pitch (J2-J4 as specified)	15 pins x 1.27mm	= 19.1mm, before adding J1, connector shrouds, or gaps between 4 bodies
At 1.0mm pitch, all four combined	17 pins x 1.0mm	= 17.0mm — equal to the entire board width with zero margin for shrouds, mounting, or the 4 separate housings

Even the best case (1.0mm pitch, everything combined into a single connector) consumes the entire board edge with nothing left for connector body overhang, mounting keep-out, or the fact that these are four separate physical parts, not one. This needs to be resolved before layout starts. Options, roughly in order of how much they preserve the rest of the spec:

- Combine J2+J4 (11 pins) into a single high-density board-to-board or FPC connector, and see if J1+J3 can share an edge with it using a taller (non-bottom-edge-only) layout.
- Move to 0.5mm pitch FPC/FFC connectors for J2 and J4, which is what this pin density realistically needs.
- Relax the 21.0 x 17.5mm constraint. It's inherited from the XIAO form factor for mechanical compatibility with an existing enclosure cavity — worth confirming whether that compatibility is a hard requirement, since a modestly larger board (even 25 x 20mm) changes this from impossible to merely tight.
- Drop J4 from this board and put the camera MCU on its own small daughter board with its own connector, if the camera link doesn't need to live on the same PCB as the Wi-Fi module.

2.4 I2C is no longer available

D4/D5 (GPIO22/GPIO23) are the stock XIAO ESP32-C6 I2C pins (SDA/SCL). This spec intentionally repurposes them for UART1 RXD_MCU and MRDY — consistent with your “Critical Update” note — which means this board variant has no native I2C breakout. Worth confirming that's acceptable for any future sensor add-ons, since it's not easily reversible without giving up UART1 pins again.

3. Pin mapping

U1 pin	Function	Direction	Connects to	Notes
EN (CHIP_PU)	Reset	in	R1 pull-up to 3.3V; SW1 to GND	unchanged from stock — do not touch
GPIO8	Strap — ROM print control	in	R3 pull-up to 3.3V	unchanged from stock — do not touch
GPIO9	Strap — boot mode select	in	R2 pull-up to 3.3V; SW2 to GND	unchanged from stock — do not touch
GPIO0, GPIO1, GPIO2	Spare (D0-D2)	—	broken out, no connect	free for future sensors/LEDs
GPIO16	UART0 TXD_DBG (D6)	out	J3 pin2	matches stock D6
GPIO7	UART0 RXD_DBG	in	J3 pin3	NOT the stock D7 pad — see §2.2
GPIO17	UART1 TXD_MCU	out	J2 pin1	this is the real D7 net — D7 pad must float, see §5
GPIO22	UART1 RXD_MCU (D4)	in	J2 pin2	sacrifices stock I2C SDA — see §2.4
GPIO3	SRDY (MCU→module wake)	in, active-low	J2 pin3	R4, 10k Ω pull-up
GPIO23	MRDY (module→MCU wake, D5)	out, active-low	J2 pin4	sacrifices stock I2C SCL — see §2.4; R5, 10k Ω pull-up
GPIO19	SPI SCK (D8)	out	J4 pin1	matches stock D8
GPIO18	SPI MOSI (D10)	out	J4 pin2	matches stock D10
GPIO20	SPI MISO (D9)	in	J4 pin3	matches stock D9
GPIO21	SPI CS (D3)	out	J4 pin4	matches stock D3

4. Net list

Net	Nodes
VBAT_IN	J1.1, D1 anode
VIN_PROT	D1 cathode, D2, C1+, U2.VIN
+3.3V	U2.VOUT, C2+, U1.VDDPST1, U1.VDDPST2, C3+, C4+, R1-R5 tops, J3.1, J4.5, TP1

Net	Nodes
GND	J1.2, D2 cathode, C1-, U2.GND, C2-, U1.GND (all), C3-, C4-, SW1/SW2 far side, J3.4, J2.5, J4.6, TP2
EN	U1.EN, R1 bottom, SW1 near side, TP3
GPIO8	U1.GPIO8, R3 bottom
GPIO9 / BOOT	U1.GPIO9, R2 bottom, SW2 near side
TXD_DBG	U1.GPIO16, J3.2
RXD_DBG	U1.GPIO7, J3.3
TXD_MCU	U1.GPIO17, J2.1
RXD_MCU	U1.GPIO22, J2.2
SRDY	U1.GPIO3, J2.3, R4 bottom
MRDY	U1.GPIO23, J2.4, R5 bottom
SCK	U1.GPIO19, J4.1
MOSI	U1.GPIO18, J4.2
MISO	U1.GPIO20, J4.3
CS	U1.GPIO21, J4.4

5. Bill of materials

Ref	Qty	Description	Value / part	Package	Example MPN	Notes
U1	1	Wi-Fi 6 / BLE5 / 802.15.4 module	ESP32-C6-MINI-1-N8	SMD module, castellated	ESP32-C6-MINI-1-N8	Unchanged from stock XIAO
U2	1	LDO regulator, 3.3V	HT7833-3.3	SOT-89-3 (confirm SOT-23-3 stock)	HT7833-3.3	Vin max 8V — see §2.1. No EN pin on this part.
D1	1	Schottky diode, reverse-polarity protection	SS12 (1A, 20V)	SMA / DO-214AC	SS12-E3/61T	Unchanged
D2	1	Bidirectional TVS, ESD/surge protection	SMBJ8.5A (was SMBJ5.0A)	SMB / DO-214AA	SMBJ8.5A-13-F	Resized for 6V system — see §2.1
C1	1	Bulk input capacitor	22µF X5R 16V	0805	GRM21BR61C226 ME44L	Voltage rating bumped for 6V input
C2	1	Bulk output capacitor	15µF X5R 16V	0805	GRM21BR61C156 ME15L	

Ref	Qty	Description	Value / part	Package	Example MPN	Notes
C3, C4	2	Decoupling, near U1 power pins	0.1μF X7R 16V	0402	GRM155R71C104 KA88D	Unchanged
R1-R5	5	Pull-up resistors	10kΩ, 1%	0402	RC0402FR-0710K L	R1 EN, R2 GPIO9, R3 GPIO8, R4 SRDY, R5 MRDY
J1	1	Battery connector, 2-pin	2.54mm pitch	THT, 2-pin	JST B2P-VH or XH-2	Confirm mates with pack's holder/connector
J2	1	Doorlock MCU connector, 5-pin	1.27mm or 2.0mm pitch	SMD header	—	Part of the 17-pin fit problem — see §2.3
J3	1	Programming connector, 4-pin	1.27mm or 2.0mm pitch	SMD header	—	Part of the 17-pin fit problem — see §2.3
J4	1	Camera MCU (SPI) connector, 6-pin	1.27mm or 2.0mm pitch	SMD header	—	Part of the 17-pin fit problem — see §2.3
SW1	1	Reset button	Momentary tactile, 3x3mm	SMD	TS-1187A-B-A-B	Unchanged from stock
SW2	1	Boot/download button	Momentary tactile, 3x3mm	SMD	TS-1187A-B-A-B	Unchanged from stock
TP1-T P3	3	Test points	3.3V, GND, EN	1mm pad or via	—	
—	—	USB-C connector	REMOVED	—	—	Omit from assembly, pads may remain unpopulated
—	—	SGM40567 battery charger IC	REMOVED	—	—	Confirmed real part on stock XIAO; omit from assembly
—	—	SGM6029C buck converter	REMOVED	—	—	Confirmed real part on stock XIAO (produces stock 3V3); omit from assembly

6. Power budget (sanity check)

- ESP32-C6 active Wi-Fi TX peak: roughly 300-450mA class — confirm against the module datasheet for your exact silicon revision.
- HT7833 500mA rating clears that peak; dropout ~500mV at 500mA means output stays regulated down to roughly 3.8V input — trivial margin at 6V nominal, only becomes relevant as 4xAA cells approach end-of-life (~4.8-5V for alkaline under load).
- Idle/quiescent draw is dominated by the HT7833's own I_q (~4-7μA) plus the chip's light-sleep current — worth a bench measurement to validate any battery-life figure you plan to publish, same caveat as the previous revision.
- D1's forward drop (Schottky, ~0.3-0.5V) subtracts from available headroom but is inconsequential at 6V input given the ~2V+ of margin above dropout.

7. Programming & bring-up procedure

- 1 Connect J3 (3.3V/TXD_DBG/RXD_DBG/GND) to a USB-UART adapter (no onboard USB-C anymore).
- 2 Hold SW2 (GPIO9/BOOT) low, briefly pulse SW1 (EN/Reset), release SW2 — standard ESP32 download-mode entry.
- 3 Flash firmware via esptool or your toolchain of choice.
- 4 Release, reset via SW1, confirm normal boot.
- 5 Connect J2 to the doorlock MCU and verify the UART1 + SRDY/MRDY Keep-Alive handshake independently of J4.
- 6 Connect J4 to the camera MCU and verify the SPI bus independently of J2, then verify both simultaneously — confirm no cross-talk given J2 and J4 are physically adjacent per §2.3.

8. Open items for next revision

- Resolve the J1-J4 connector pin count / pitch / footprint problem — see §2.3. This blocks layout.
- Decide battery chemistry (alkaline/NiMH only, or re-qualify for lithium AA) and put it on the packaging — see §2.1.
- Confirm the D7 castellation is left floating (not accidentally tied to GPIO17/TXD_MCU) — see §2.2.
- Confirm SOT-23-3 HT7833-3.3 stock availability if SOT-89 doesn't suit the layout — see §5.
- Confirm no future need for I2C before finalizing — see §2.4.
- Re-verify J2/J4 signal integrity given proximity to the door motor, per your own note about wanting a dedicated GND on J2.